

Design a Swing-based GUI application for a hospital to manage patient appointments, including booking, updating, and canceling schedules. Use AWT/Swing controls such as buttons, text fields, tables, and menus to facilitate user interaction. Implement event listeners to handle actions like appointment confirmation and deletion. Ensure the system can handle dynamic patient data efficiently. Evaluate different layout managers (FlowLayout, GridLayout, BorderLayout) for organizing complex UI components. Analyze how the design supports scalability when the number of patients increases. Propose improvements for better usability and real-time updates.

Code

```
import javax.swing.*;
import javax.swing.table.*;
import java.awt.*;
import java.awt.event.*;
import java.util.ArrayList;

public class HospitalAppointmentSystem extends JFrame {

    static class Appointment {
        static int counter = 1;
        int id; String patient, doctor, date, time, status;
        Appointment(String patient, String doctor, String date, String time) {
            this.id = counter++; this.patient = patient;
            this.doctor = doctor; this.date = date;
            this.time = time; this.status = "Confirmed";
        }
        Object[] toRow() {
            return new Object[]{id, patient, doctor, date, time, status};
        }
    }

    private final ArrayList<Appointment> appointments = new ArrayList<>();
    private final DefaultTableModel tableModel;
    private final JTable table;
    private final JTextField tfPatient = new JTextField(15);
    private final JTextField tfDoctor = new JTextField(15);
    private final JTextField tfDate = new JTextField(12);
    private final JTextField tfTime = new JTextField(10);
    private final JTextField tfSearch = new JTextField(18);
    private final JLabel statusBar = new JLabel(" Ready");

    public HospitalAppointmentSystem() {
        super("Hospital Appointment Management System");
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setSize(950, 620);
        setLocationRelativeTo(null);
        String[] cols = {"ID", "Patient Name", "Doctor", "Date", "Time", "Status"};
        tableModel = new DefaultTableModel(cols, 0) {
            public boolean isCellEditable(int r, int c) { return false; }
        };
        table = new JTable(tableModel);
        styleTable();
        setLayout(new BorderLayout(6, 6));
        add(buildHeader(), BorderLayout.NORTH);
        add(buildFormPanel(), BorderLayout.WEST);
        add(buildTablePanel(), BorderLayout.CENTER);
        add(buildStatusBar(), BorderLayout.SOUTH);
        seedData();
        setVisible(true);
    }

    private JPanel buildHeader() {
        JPanel p = new JPanel(new FlowLayout(FlowLayout.LEFT));
        p.setBackground(new Color(26, 82, 118));
        JLabel lbl = new JLabel(" Hospital Appointment Management System");
        lbl.setFont(new Font("Segoe UI", Font.BOLD, 18));
        lbl.setForeground(Color.WHITE);
        p.add(lbl); return p;
    }

    private JPanel buildFormPanel() {
        JPanel wrap = new JPanel(new BorderLayout());
        wrap.setBorder(BorderFactory.createEmptyBorder(8, 8, 8, 4));
        wrap.setPreferredSize(new Dimension(260, 0));
        JPanel form = new JPanel(new GridLayout(0, 1, 4, 4));
        form.setBorder(BorderFactory.createTitledBorder("Book / Update Appointment"));
        form.add(label("Patient Name *")); form.add(tfPatient);
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        form.add(label("Doctor *"));          form.add(tfDoctor);
        form.add(label("Date (DD-MM-YYYY) *")); form.add(tfDate);
        form.add(label("Time (HH:MM) *"));   form.add(tfTime);
        JButton btnBook   = btn("Book",    new Color(39,174,96),   Color.WHITE);
        JButton btnUpdate = btn("Update",  new Color(41,128,185),  Color.WHITE);
        JButton btnClear  = btn("Clear",   new Color(127,140,141),Color.WHITE);
        form.add(btnBook); form.add(btnUpdate); form.add(btnClear);
        JPanel srch = new JPanel(new GridLayout(0,1,4,4));
        srch.setBorder(BorderFactory.createTitledBorder("Search"));
        srch.add(label("Patient / Doctor name")); srch.add(tfSearch);
        JButton btnSearch = btn("Search",  new Color(142,68,173), Color.WHITE);
        JButton btnReset  = btn("Reset",   new Color(127,140,141),Color.WHITE);
        srch.add(btnSearch); srch.add(btnReset);
        JButton btnCancel = btn("Cancel Selected", new Color(192,57,43), Color.WHITE);
        wrap.add(form, BorderLayout.NORTH);
        wrap.add(srch, BorderLayout.CENTER);
        JPanel bot = new JPanel(new BorderLayout());
        bot.add(btnCancel, BorderLayout.NORTH);
        wrap.add(bot, BorderLayout.SOUTH);
        btnBook.addActionListener(e -> bookAppointment());
        btnUpdate.addActionListener(e -> updateAppointment());
        btnClear.addActionListener(e -> clearForm());
        btnSearch.addActionListener(e -> searchAppointments());
        btnReset.addActionListener(e -> { tfSearch.setText(""); refreshTable(appointments); });
        btnCancel.addActionListener(e -> cancelSelected());
        table.getSelectionModel().addListSelectionListener(ev -> {
            if (!ev.getValueIsAdjusting()) populateForm();
        });
        return wrap;
    }

    private JPanel buildTablePanel() {
        JPanel p = new JPanel(new BorderLayout(4,4));
        p.setBorder(BorderFactory.createTitledBorder("Appointment Schedule"));
        p.add(new JScrollPane(table), BorderLayout.CENTER);
        JPanel stats = new JPanel(new FlowLayout(FlowLayout.LEFT));
        JLabel lTotal = new JLabel();
        JLabel lConf  = new JLabel();
        JLabel lCanc  = new JLabel();
        stats.add(lTotal); stats.add(lConf); stats.add(lCanc);
        tableModel.addTableModelListener(e -> {
            long conf = appointments.stream()
                .filter(a -> a.status.equals("Confirmed")).count();
            long canc = appointments.stream()
                .filter(a -> a.status.equals("Cancelled")).count();
            lTotal.setText(" Total: " + appointments.size() + " |");
            lConf.setText(" Confirmed: " + conf + " |");
            lCanc.setText(" Cancelled: " + canc);
        });
        p.add(stats, BorderLayout.SOUTH);
        return p;
    }
}

```

Expected Output / GUI Description

Hospital Appointment Management System

Book / Update Appointment

Patient Name *
Sunita Verma

Doctor *
Dr. Mehta

Date (DD-MM-YYYY) *
20-08-2025

Time (HH:MM) *
11:30

Book
Update
Clear

Search

Patient / Doctor name
Type to search...

Search
Reset

Appointment Schedule 4 records

ID	Patient Name	Doctor	Date	Time	Status
1	Ravi Kumar	Dr. Sharma	10-08-2025	09:00	Confirmed
2	Priya Patel	Dr. Sharma	11-08-2025	10:30	Confirmed
3					Cancelled
4	Sunita Verma	Dr. Sharma	13-08-2025	14:00	Confirmed

Total: 4 Confirmed: 3 Cancelled: 1